

第五章 移动机器人感知技术

Kalman Filter Based Localization

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主要内容

Introduction

Motivation

Definitions

Bayes-based localization

Introduction

Probability theory

Description of the probabilistic localization problem

Kalman Filter based Localization

Principle of Kalman Filter

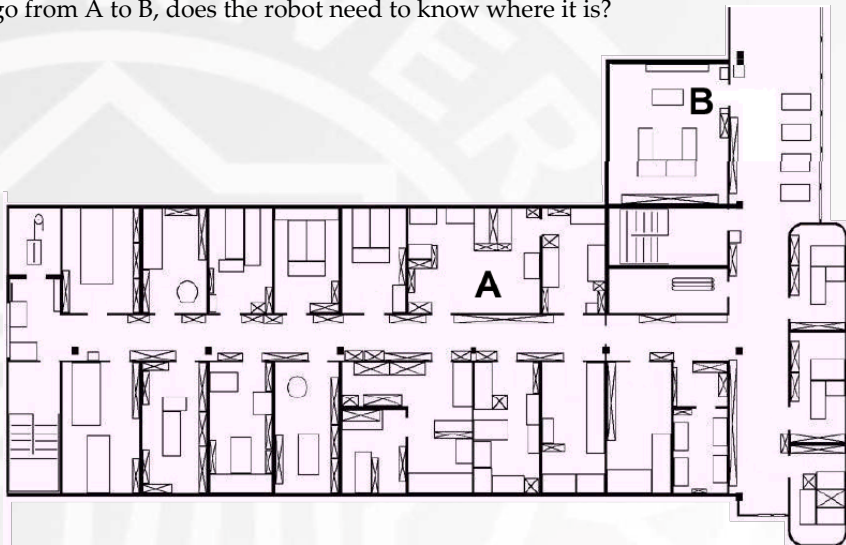
优化策略角度下的卡尔曼滤波算法

Strategies of Kalman Filter

Kalman Filter for Mobile Robot

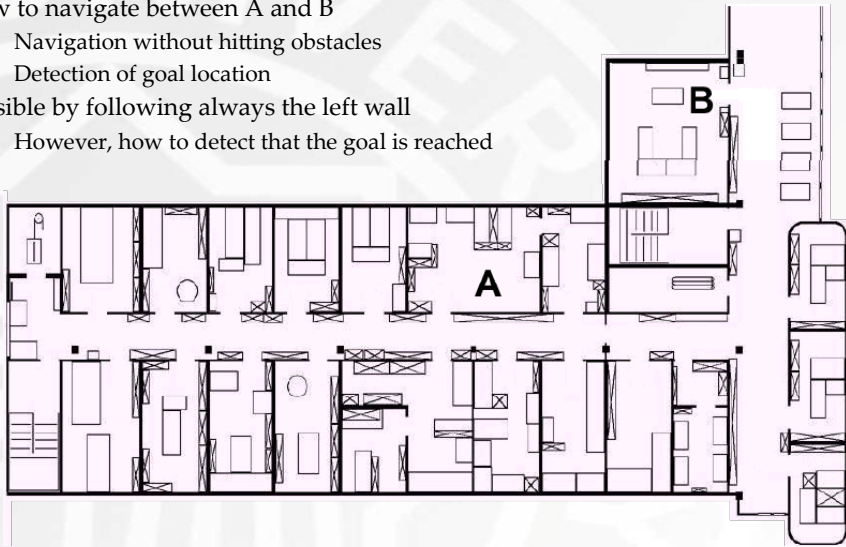
Introduction

- To go from A to B, does the robot need to know where it is?



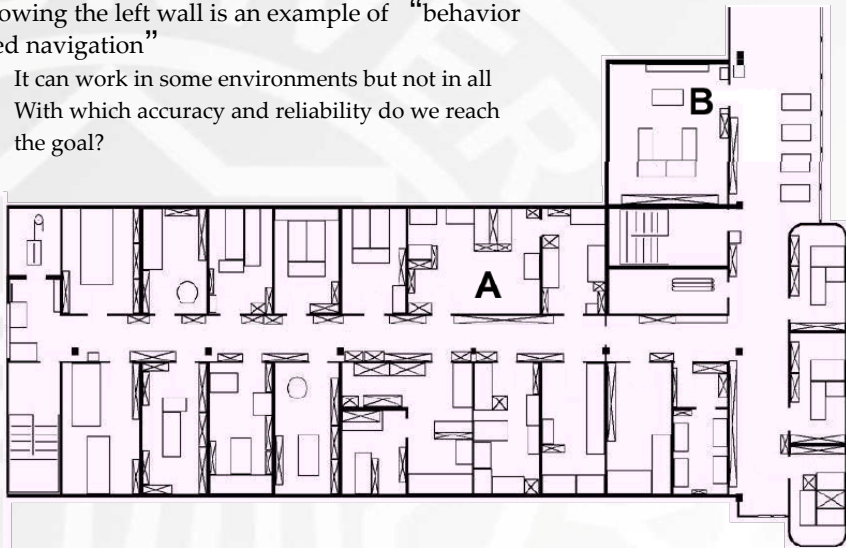
Introduction

- How to navigate between A and B
 - Navigation without hitting obstacles
 - Detection of goal location
- Possible by following always the left wall
 - However, how to detect that the goal is reached



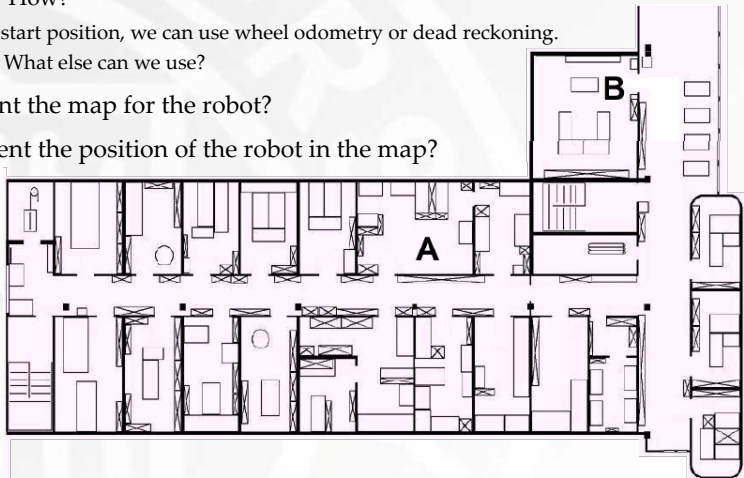
Introduction

- Following the left wall is an example of “behavior based navigation”
 - It can work in some environments but not in all
 - With which accuracy and reliability do we reach the goal?



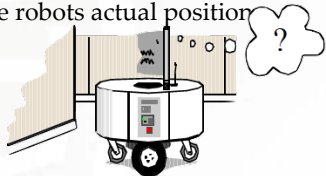
Introduction

- As opposed to behavior based navigation is “map based navigation”
 - Assuming that the map is known, at every time step the robot has to know where it is. How?
 - If we know the start position, we can use wheel odometry or dead reckoning.
Is this enough? What else can we use?
- But how do we represent the map for the robot?
- And how do we represent the position of the robot in the map?



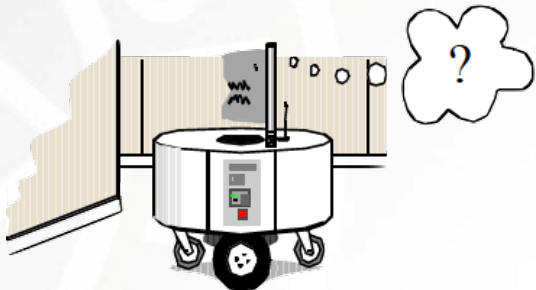
The problem

- Consider a mobile robot moving in a known environment.
- As it starts to move, say from a precisely known location, it might keep track of its location using odometry.
- However, after a certain movement the robot will get very uncertain about its position
⇒ update using an observation of its environment
- Observation leads also to an estimate of the robot position which can then be fused with the odometric estimation to get the best possible update of the robots actual position



Definitions

- Global Localization
 - The robot is not told its initial position
 - Its position must be estimated from scratch
- Position Tracking
 - A robot knows its initial position and “only” has to accommodate small errors in its odometry as it moves
- Kidnapped Robot Problem





How to localize?

- Localization based on external sensors, beacons or landmarks
- Odometry
- Map Based Localization - without external sensors or artificial landmarks, just use robot onboard sensors
 - Example: Probabilistic Map Based Localization



Basic Concepts

- Probability theory $\Rightarrow$ error propagation, sensor fusion
- Belief representation(map/position) $\Rightarrow$ discrete / continuous
- Motion model $\Rightarrow$ odometry model
- Sensing $\Rightarrow$ measurement model



Probability theory

Discrete probability distributions:

$$p(x) \in [0, 1] \text{ for all } x \in \Omega \quad \text{and} \quad \sum_{x \in \Omega} p(x) = 1.$$

Continuous probability distributions:

$$F(x) = P(X \leq x)$$

F is a monotonically non-decreasing, right-continuous function.

Probability Density Function(PDF) $p(x)$:

$$p(x) = \frac{dF(x)}{dx} \quad \text{and} \quad \int_x p(x) dx$$



Probability theory

- The Gaussian distribution(the normal distribution):

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\mu = E[X] = \int_{-\infty}^{\infty} xp(x)dx$$

$$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 p(x)dx$$

- Joint distribution function $p(x, y)$
- Conditional probability $p(x|y)$
- If two random variables X_1 and X_2 are independent, then

$$E[X_1 X_2] = E[X_1]E[X_2]$$

$$Val[X_1 + X_2] = Val[X_1] + Val[X_2]$$

$$p(x, y) = p(x)p(y) \quad \text{and} \quad p(x, y) = p(x)p(y)$$



Probability theory

- The Gaussian distribution(the normal distribution):

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- The expectation $E(x)$ and the Variance $Var(X)$

$$\mu = E[X] = \int_{-\infty}^{\infty} xp(x)dx \quad \text{and} \quad Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 p(x)dx$$

- Joint distribution function $p(x, y)$
- Conditional probability $p(x|y)$:

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

- If two random variables X_1 and X_2 are **independent**, then



Probability theory

- The Gaussian distribution(the normal distribution):

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- The expectation $E(x)$ and the Variance $Val(X)$

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- Joint distribution function $p(x, y)$
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$$p(x|y) = \frac{p(x, y)}{p(y)}$$

- If two random variables X_1 and X_2 are **independent**, then

$$E[X_1 X_2] = E[X_1]E[X_2] \quad \text{and} \quad Val[X_1 + X_2] = Val[X_1] + Val[X_2]$$

$$p(x, y) = p(x)p(y) \quad \text{and} \quad p(x|y) = p(x)$$



Probability theory

- Law of total probability:

$$p(x) = \sum_y p(x|y)p(y) \quad \text{and} \quad p(x) = \int_y p(x|y)p(y)dy$$

- Bayes's theorem:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

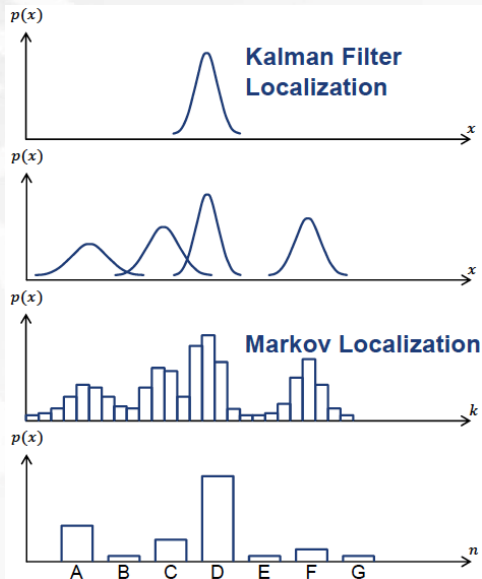
注意:

贝叶斯定理的核心思想:

新信息出现后的 A 的概率 = A 的概率 $\times$ 新信息带来的调整

Belief Representation

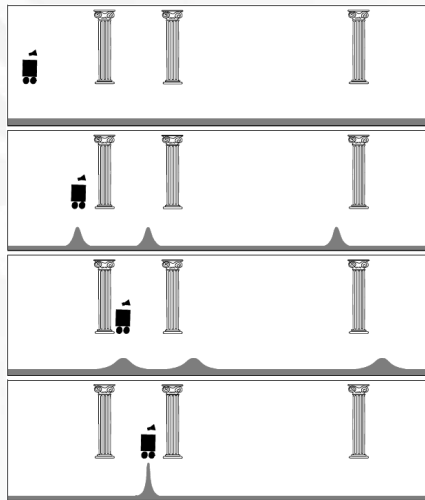
- Continuous map with single hypothesis probability distribution $p(x)$
- Continuous map with multiple hypotheses probability distribution $p(x)$
- Discretized metric map (grid k) with probability distribution $p(x)$
- Discretized topological map (nodes n) with probability distribution $p(x)$



Map-based localization

Improving belief state by moving:

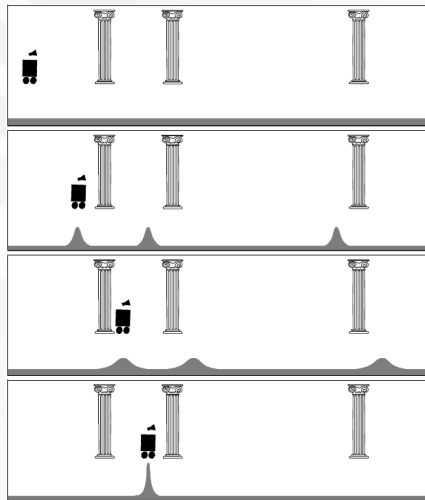
- The robot is placed somewhere in the environment but it is not told its location



Map-based localization

Improving belief state by moving:

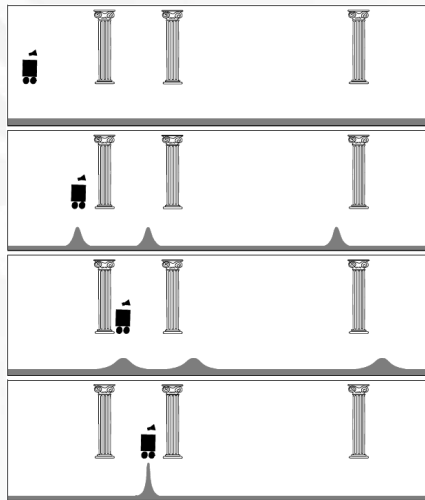
- The robot is placed somewhere in the environment but it is not told its location
- The robot queries its sensors and finds it is next to a pillar



Map-based localization

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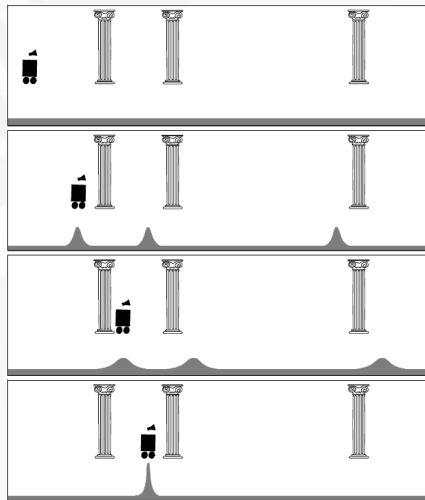
- The robot is placed somewhere in the environment but it is not told its location
- The robot queries its sensors and finds it is next to a pillar
- The robot moves one meter forward. To account for inherent noise in robot motion the new belief is smoother



Map-based localization

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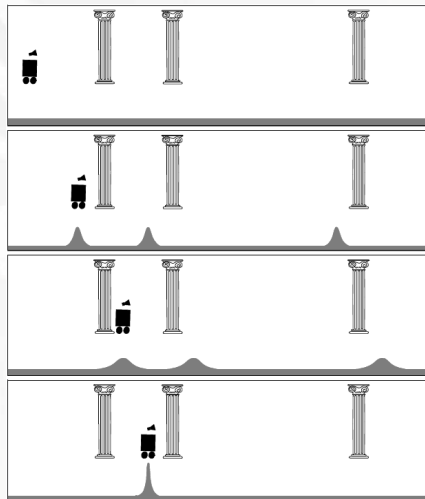
- The robot is placed somewhere in the environment but it is not told its location
- The robot queries its sensors and finds it is next to a pillar
- The robot moves one meter forward. To account for inherent noise in robot motion the new belief is smoother
- The robot queries its sensors and again it finds itself next to a pillar



Map-based localization

Improving belief state by moving:

- The robot is placed somewhere in the environment but it is not told its location
- The robot queries its sensors and finds it is next to a pillar
- The robot moves one meter forward. To account for inherent noise in robot motion the new belief is smoother
- The robot queries its sensors and again it finds itself next to a pillar
- Finally, it updates its belief by combining this information with its previous belief



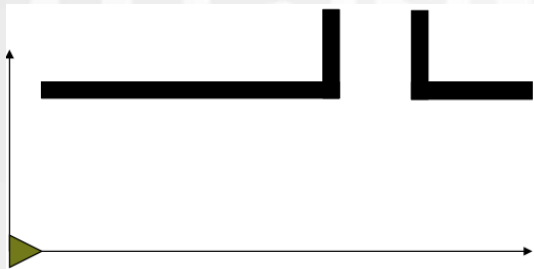
Why a probabilistic approach for mobile robot localization?

- Because the data coming from the robot sensors are affected by measurement errors, we can only compute the **probability that the robot is in a given configuration**.
- The key idea in probabilistic robotics is to represent uncertainty using probability theory: instead of giving a single best estimate of the current robot configuration, probabilistic robotics represents the robot configuration as a **probability distribution over the all possible robot poses**. This probability is called **belief**.



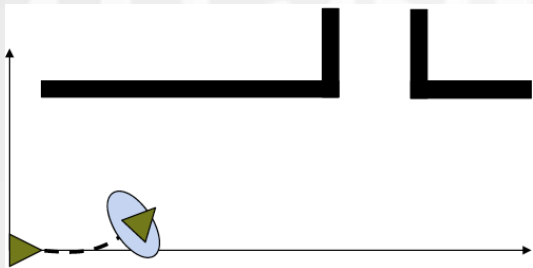
Description of the probabilistic localization problem

- Consider a mobile robot moving in a known environment.



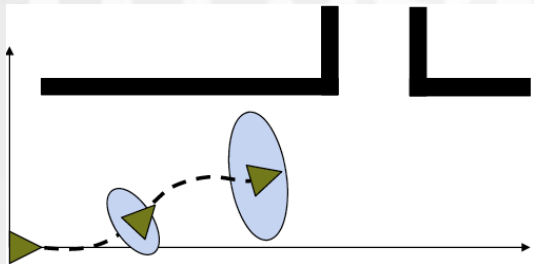
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- Consider a mobile robot moving in a known environment.
- As it starts to move, say from a precisely known location, it can keep track of its motion using odometry.



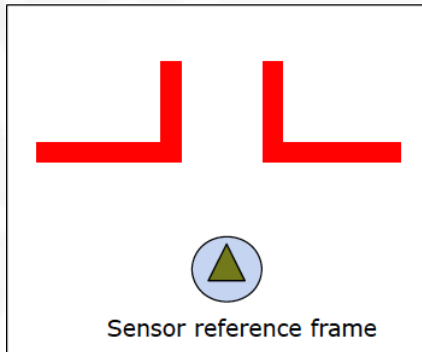
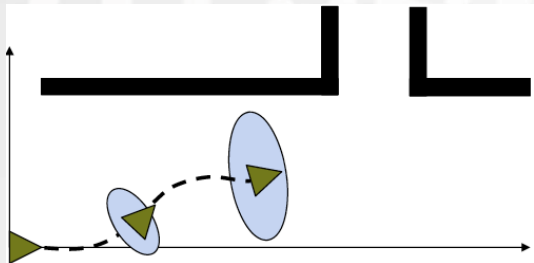
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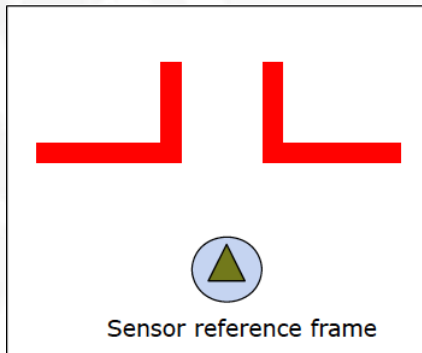
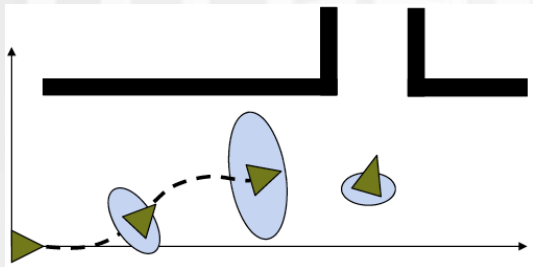
Description of the probabilistic localization problem

- Consider a mobile robot moving in a known environment.
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Description of the probabilistic localization problem

- Consider a mobile robot moving in a known environment.
- As it starts to move, say from a precisely known location, it can keep track of its motion using odometry.
- The robot makes an observation and updates its position and uncertainty





Characteristics

- Continuous

- Precision bound by sensor data
- Typically single hypothesis pose estimate
- Lost when diverging (for single hypothesis)
- Compact representation and typically reasonable in processing power.

- Discrete

- Precision bound by resolution of discretisation
- Typically multiple hypothesis pose estimate
- Never lost (when diverges converges to another cell)
- Important memory and processing power needed. (not the case for topological maps)

Rudolf Emil Kalman

鲁道夫·卡尔曼

匈牙利裔美国数学家，1930 年出生于匈牙利首都布达佩斯。1953 年于麻省理工学院获得电机工程学士，翌年硕士学位。1957 年于哥伦比亚大学获得博士学位。1964 年至 1971 年任职斯坦福大学。1971 年至 1992 年任佛罗里达大学数学系统理论中心（Center for Mathematical System Theory）主任。1972 起任瑞士苏黎世联邦理工学院数学系统理论中心主任直至退休。

- 美国科学院、美国工程院院士
- 匈牙利、法国、俄罗斯外籍院士
- IEEE 荣誉奖章
- IEEE 百年奖章





What is a Kalman Filter?

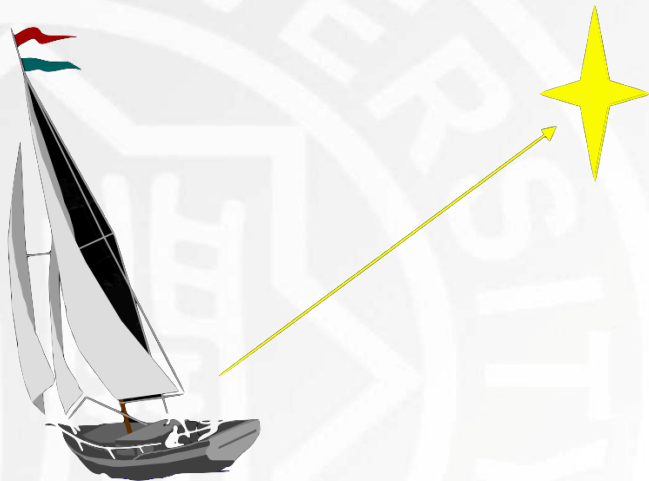
- Just some applied math.
- A linear system: $f(a + b) = f(a) + f(b)$.
- Noisy data in $\Rightarrow$ hopefully less noisy out.
- But delay is the price for filtering
- Pure KF does not even adapt to the data.



What is it used for?

- Tracking missiles
- Tracking heads/hands/drumsticks
- Extracting lip motion from video
- Fitting Bezier patches to point data
- Lots of computer vision applications
- Economics
- Navigation

A really simple example



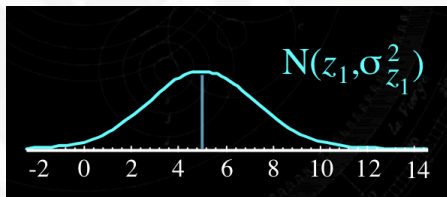
Gary makes a measurement

$$z_1, \sigma_{z_1}^2$$

$$\hat{x}_1 = z_1$$

$$\hat{\sigma}_1^2 = \sigma_{z_1}^2$$

Conditional Density Function



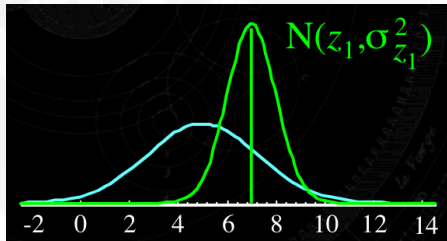
Greg makes a measurement

$$z_2, \sigma_{z_2}^2$$

$$\hat{x}_2 = \dots ?$$

$$\hat{\sigma}_2^2 = \dots ?$$

Conditional Density Function



注意:

Not all the difference is error!



应用贝叶斯定理

- 令 q 表示测量位置, $p_1(q)$ 、 $p_2(q)$ 分别表示两次**独立**的测量结果的信任度, 假设这两次测量结果均服从正态分布:

$$p_1(q) = N(\hat{q}_1, \sigma_1^2) \quad p_2(q) = N(\hat{q}_2, \sigma_2^2)$$



应用贝叶斯定理

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$$p_1(q) = N(\hat{q}_1, \sigma_1^2) \quad p_2(q) = N(\hat{q}_2, \sigma_2^2)$$

- 由全概率定理可知, 两次测量之后, 最终分布 $p(q)$ 也服从正态分布:

$$p(q) = N(\hat{q}, \sigma^2)$$

且

$$p(q) = p_1(q) \cdot p_2(q) = \frac{1}{\sigma_1 \sqrt{2\pi}} \exp\left(-\frac{(q - \hat{q}_1)^2}{2\sigma_1^2}\right) \cdot \frac{1}{\sigma_2 \sqrt{2\pi}} \exp\left(-\frac{(q - \hat{q}_2)^2}{2\sigma_2^2}\right)$$

应用贝叶斯定理

- 令 q 表示测量位置, $p_1(q)$ 、 $p_2(q)$ 分别表示两次**独立**的测量结果的信任度, 假设这两次测量结果均服从正态分布:

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且

$$\begin{aligned} p(q) &= p_1(q) \cdot p_2(q) = \frac{1}{\sigma_1 \sqrt{2\pi}} \exp\left(-\frac{(q - \hat{q}_1)^2}{2\sigma_1^2}\right) \cdot \frac{1}{\sigma_2 \sqrt{2\pi}} \exp\left(-\frac{(q - \hat{q}_2)^2}{2\sigma_2^2}\right) \\ &= \frac{1}{2\sigma_1 \sigma_2 \pi} \exp\left(-\frac{(q - \hat{q}_1)^2}{2\sigma_1^2} - \frac{(q - \hat{q}_2)^2}{2\sigma_2^2}\right) \end{aligned}$$



应用贝叶斯定理

由于:

$$\exp \left(-\frac{(q - \hat{q}_1)^2}{2\sigma_1^2} - \frac{(q - \hat{q}_2)^2}{2\sigma_2^2} \right)$$



应用贝叶斯定理

由于:

$$\begin{aligned} & \exp \left(-\frac{(q - \hat{q}_1)^2}{2\sigma_1^2} - \frac{(q - \hat{q}_2)^2}{2\sigma_2^2} \right) \\ &= \exp \left(-\frac{1}{2} \left(\frac{q^2(\sigma_1^2 + \sigma_2^2) - 2q(\hat{q}_1\sigma_2^2 + \hat{q}_2\sigma_1^2) + (\hat{q}_1^2\sigma_2^2 + \hat{q}_2^2\sigma_1^2)}{\sigma_1^2\sigma_2^2} \right) \right) \end{aligned}$$

应用贝叶斯定理

由于:

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应用贝叶斯定理

由于:

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相对于变量 q 为常数



融合结果

- $p(q)$ 的分布可写成如下形式:

$$p(q) = N(\hat{q}, \sigma^2) = \Omega \exp \left(-\frac{1}{2} \frac{\left(q - \frac{\hat{q}_1 \sigma_2^2 + \hat{q}_2 \sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right)^2}{\frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}} \right)$$

融合结果

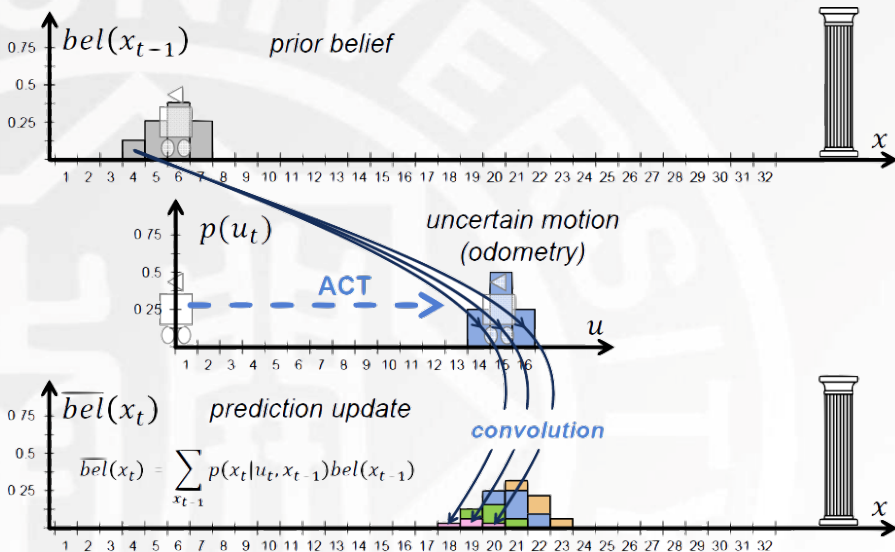
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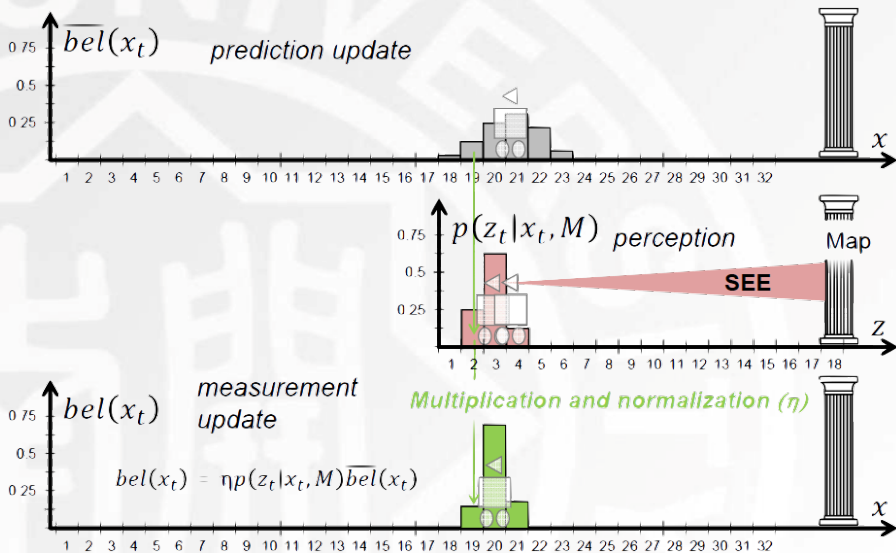
- 可得:

$$\hat{q} = \frac{\hat{q}_1 \sigma_2^2 + \hat{q}_2 \sigma_1^2}{\sigma_1^2 + \sigma_2^2} \Leftrightarrow \hat{q} = \frac{\frac{1}{\sigma_1^2} \hat{q}_1 + \frac{1}{\sigma_2^2} \hat{q}_2}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} \Leftrightarrow \hat{q} = \hat{q}_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (\hat{q}_2 - \hat{q}_1)$$
$$\sigma^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} \Leftrightarrow \frac{1}{\sigma^2} = \frac{\sigma_1^2 + \sigma_2^2}{\sigma_1^2 \sigma_2^2} = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \Leftrightarrow \sigma^2 = \sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2 + \sigma_2^2}$$

ACT | using motion model and its uncertainties



SEE | estimation of position based on perception and map





一般形式下的融合结果

- 令 $q \in R^n$ 表示系统状态, $p_1(q)$ 、 $p_2(q)$ 分别表示两次**独立**的测量结果的信任度, 假设这两次测量结果均服从正态分布:

$$p_1(q) = N(\hat{q}_1, P) \quad p_2(q) = N(\hat{q}_2, Q)$$

其中, P 、 Q 分别表示 $p_1(q)$ 和 $p_2(q)$ 所对应的协方差阵。



一般形式下的融合结果

- 令 $q \in R^n$ 表示系统状态, $p_1(q)$ 、 $p_2(q)$ 分别表示两次**独立**的测量结果的信任度, 假设这两次测量结果均服从正态分布:

$$p_1(q) = N(\hat{q}_1, P) \quad p_2(q) = N(\hat{q}_2, Q)$$

其中, P 、 Q 分别表示 $p_1(q)$ 和 $p_2(q)$ 所对应的协方差阵。

- 由全概率定理可知, 两次测量之后, 最终分布 $p(q)$ 也服从正态分布:

$$p(q) = N(\hat{q}, R)$$

由上面标量融合结果可得,

$$\hat{q} = \hat{q}_1 + P(P + Q)^{-1}(\hat{q}_2 - \hat{q}_1) \Leftrightarrow \hat{q} = \hat{q}_1 + K(\hat{q}_2 - \hat{q}_1)$$

$$R = P - P(P + Q)^{-1}P \Leftrightarrow R = P - K \cdot \Sigma_{IN} \cdot K^T$$

其中,

$$K = P(P + Q)^{-1}(\text{卡尔曼增益}) \quad \Sigma_{IN} = P + Q(\text{创新协方差})$$



优化策略角度下的卡尔曼滤波算法

问题描述:

一线性离线系统可用如下方程描述:

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1} \quad (1)$$

$$z_k = Hx_k + v_k \quad (2)$$

其中 $x_k \in R^n$ 为系统状态, $z_k \in R^m$ 为测量输出, $u_k \in R^l$ 为系统输入。 $w_k \in R^n$ 为过程噪声 (Process Noise), $v_k \in R^m$ 为测量噪声 (Measurement Noise) 为白噪音:

$$p(w) = N(0, Q) \quad p(v) = N(0, R) \quad (3)$$

其中, Q 为过程噪音协方差阵, R 为测量噪声协方差阵

求取通过系统状态转移方程预测的系统状态 x_k 和通过测量方程观测到的系统状态 z_k 下最优的状态估计。

相关概念

先验估计、先验估计误差及先验估计方差 $\hat{x}_k^-$ 、 e_k^- 、 P_k^- ：

- Define $\hat{x}_k^- \in R^n$ (note the "super minus") to be a *priori state estimate* at step k given knowledge of the process transform equation by the state prior x_{k-1} prior to step k .
- Define a *priori estimate errors* as $e_k^- \equiv x_k - \hat{x}_k^-$.
- The *priori estimate error covariance* is then defined as $P_k^- = E[e_k^- e_k^{-T}]$.

后验估计、后验估计误差及后验估计方差 $\hat{x}_k$ 、 e_k 、 P_k ：

- Define $\hat{x}_k \in R^n$ to be a *posteriori state estimate* at step k combined the measurement $z_k(k)$ with the priori state estimate $\hat{x}_k^-$.
- Define a *posteriori estimate errors* as $e_k \equiv x_k - \hat{x}_k$.
- The *posteriori estimate error covariance* is then defined as $P_k = E[e_k e_k^T]$.



相关概念

In deriving the equations for the Kalman filter, we begin with the goal of finding an equation that computes an a *posteriori* state estimate $\hat{x}$ as a linear combination of a *priori* estimate $\hat{x}_k^-$ and a weighted difference between an actual measurement z_k and a measurement prediction $H\hat{x}_k^-$ as shown below:

$$\hat{x}_k = \hat{x}_k^- + K(z_k - H\hat{x}_k^-) \quad (4)$$

where

- The *gain* or *blending factor*: $K \in R^{n \times m}$
- The *measurement innovation*, or the *residual*: $(z_k - H\hat{x}_k^-)$

优化策略角度下的卡尔曼滤波算法的目标函数：

$$\arg \min_K P_k \quad (5)$$



推导过程

$$P_k = E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T]$$



推导过程

$$\begin{aligned} P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\ &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \end{aligned}$$



推导过程

$$\begin{aligned} P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\ &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\ &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \end{aligned}$$



推导过程

$$\begin{aligned} P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\ &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\ &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\ &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \end{aligned}$$



推导过程

$$\begin{aligned} P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\ &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\ &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\ &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \\ &= E[(e_k^- - K(He_k^- + v_k))(e_k^- - K(He_k^- + v_k))^T] \end{aligned}$$



推导过程

$$\begin{aligned} P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\ &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\ &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\ &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \\ &= E[(e_k^- - K(He_k^- + v_k))(e_k^- - K(He_k^- + v_k))^T] \\ &= E[((I - KH)e_k^- - Kv_k)((I - KH)e_k^- - Kv_k)^T] \end{aligned}$$



推导过程

$$\begin{aligned}
 P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\
 &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \\
 &= E[(e_k^- - K(He_k^- + v_k))(e_k^- - K(He_k^- + v_k))^T] \\
 &= E[((I - KH)e_k^- - Kv_k)((I - KH)e_k^- - Kv_k)^T] \\
 &= E[(I - KH)e_k^- e_k^{-T} (I - KH)^T + Kv_k v_k^T K^T - Kv_k e_k^{-T} (I - KH)^T \\
 &\quad - (I - KH)e_k^- v_k^T K^T]
 \end{aligned}$$

推导过程

$$\begin{aligned}
 P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\
 &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \\
 &= E[(e_k^- - K(He_k^- + v_k))(e_k^- - K(He_k^- + v_k))^T] \\
 &= E[((I - KH)e_k^- - Kv_k)((I - KH)e_k^- - Kv_k)^T] \\
 &= E[(I - KH)e_k^- e_k^{-T} (I - KH)^T + Kv_k v_k^T K^T - Kv_k e_k^{-T} (I - KH)^T \\
 &\quad - (I - KH)e_k^- v_k^T K^T] \\
 &= (I - KH) \underbrace{E[e_k^- e_k^{-T}]}_{P_k^-} (I - KH)^T + K \underbrace{E[v_k v_k^T]}_R K^T - K \underbrace{E[v_k e_k^{-T}]}_0 (I - KH)^T \\
 &\quad - (I - KH) \underbrace{E[e_k^- v_k^T]}_0 K^T
 \end{aligned}$$

推导过程

$$\begin{aligned}
 P_k &= E[e_k e_k^T] = E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \\
 &= E[\{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\} \{x_k - [\hat{x}_k^- + K(z_k - H\hat{x}_k^-)]\}^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-)) ((x_k - \hat{x}_k^-) - K(Hx_k + v_k - H\hat{x}_k^-))^T] \\
 &= E[((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k)) ((x_k - \hat{x}_k^-) - K(H(x_k - \hat{x}_k^-) + v_k))^T] \\
 &= E[(e_k^- - K(He_k^- + v_k))(e_k^- - K(He_k^- + v_k))^T] \\
 &= E[((I - KH)e_k^- - Kv_k)((I - KH)e_k^- - Kv_k)^T] \\
 &= E[(I - KH)e_k^- e_k^{-T} (I - KH)^T + Kv_k v_k^T K^T - Kv_k e_k^{-T} (I - KH)^T \\
 &\quad - (I - KH)e_k^- v_k^T K^T] \\
 &= (I - KH) \underbrace{E[e_k^- e_k^{-T}]}_{P_k^-} (I - KH)^T + K \underbrace{E[v_k v_k^T]}_R K^T - K \underbrace{E[v_k e_k^{-T}]}_0 (I - KH)^T \\
 &\quad - (I - KH) \underbrace{E[e_k^- v_k^T]}_0 K^T \\
 &= (I - KH) P_k^- (I - KH)^T + K R K^T
 \end{aligned}$$

最优值求取

目标函数：

$$\arg \min_K P_k = (I - KH)P_k^-(I - KH)^T + KRK^T$$

可得最优的卡尔曼增益如下：

$$\frac{\partial P_k}{\partial K} = -2P_k^-H^T + 2KHP_k^-H^T + 2KR = 0 \Rightarrow K = P_k^-H^T(H P_k^-H^T + R)^{-1}$$

注意：

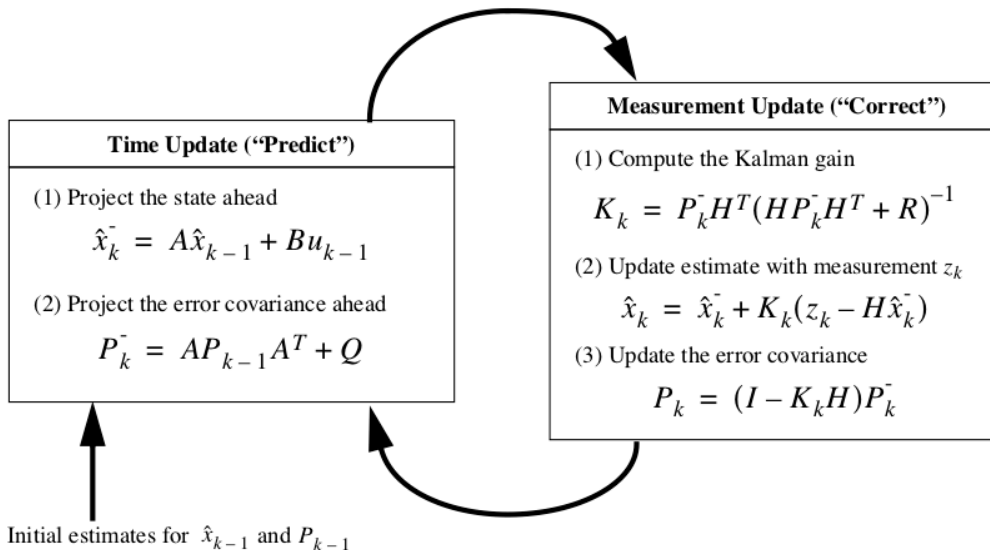
- When the measurement error covariance R approaches zero,

$$\lim_{R \rightarrow 0} K = H^{-1}$$

- When the a priori estimate error covariance P_k^- approaches zero,

$$\lim_{P_k^- \rightarrow 0} K = 0$$

Strategies of Kalman Filter



Measurement Model

“What you see from where you are”

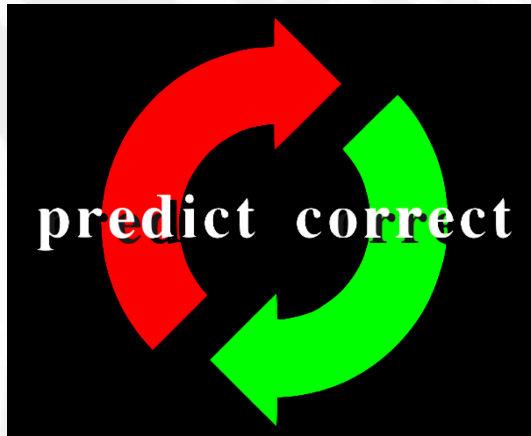
not

“Where you are from what you see”

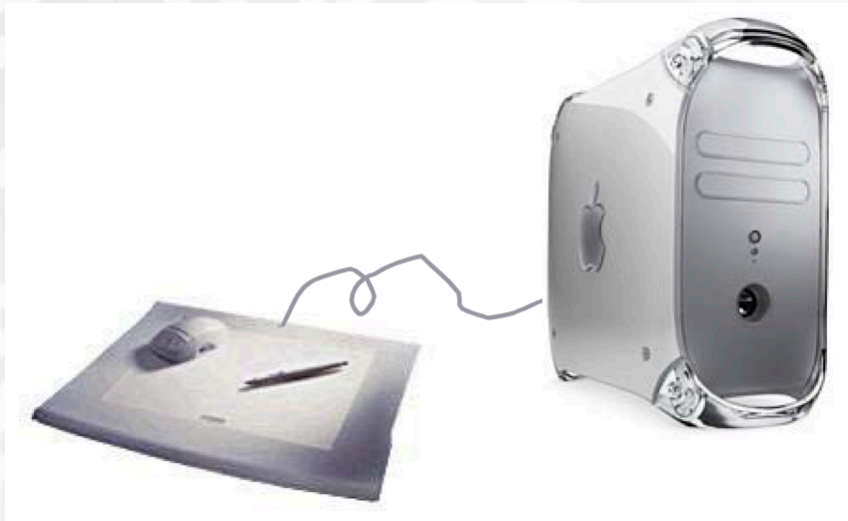
Predict → Correct

KF operates by

- Predicting the new state and its uncertainty
- Correcting with the new measurement



Example: 2D Position-Only





Process Model

$$\begin{bmatrix} x_k \\ y_k \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_{k-1} \\ y_{k-1} \end{bmatrix} + \begin{bmatrix} \tilde{x}_{k-1} \\ \tilde{y}_{k-1} \end{bmatrix}$$

State $\bar{x}_k$ State Transition A State $\bar{x}_{k-1}$ Noise $\bar{w}_{k-1}$

$$\bar{x}_k = A\bar{x}_{k-1} + \bar{w}_{k-1}$$



Measurement Model

$$\begin{bmatrix} u_k \\ v_k \end{bmatrix} = \begin{bmatrix} H_x & 0 \\ 0 & H_y \end{bmatrix} \begin{bmatrix} x_k \\ y_k \end{bmatrix} + \begin{bmatrix} \tilde{u}_k \\ \tilde{v}_k \end{bmatrix}$$

Measurement $\bar{z}_k$

Measurement Matrix H

State $\bar{x}_k$

Noise $\bar{v}_k$

$$\bar{z}_k = H\bar{x}_k + \bar{v}_k$$



Preparation

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

State
Transition

$$Q = E [\bar{w} \times \bar{w}^T] = \begin{bmatrix} q_{xx} & 0 \\ 0 & q_{yy} \end{bmatrix}$$

Process
Noise
Covariance

$$R = E [\bar{v} \times \bar{v}^T] = \begin{bmatrix} r_{xx} & 0 \\ 0 & r_{yy} \end{bmatrix}$$

Measurement
Noise
Covariance



Initialization

$$\bar{x}_0 = H \bar{z}_0$$
$$P_0 = \begin{bmatrix} v_{x0} & 0 \\ 0 & v_{y0} \end{bmatrix}$$



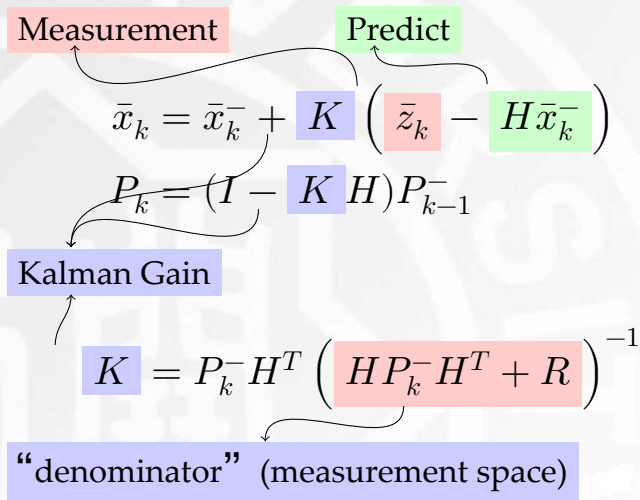
Predict

$$\begin{aligned}\bar{x}_k^- &= A \bar{x}_{k-1} \\ P_k^- &= A P_{k-1} A^T + Q\end{aligned}$$

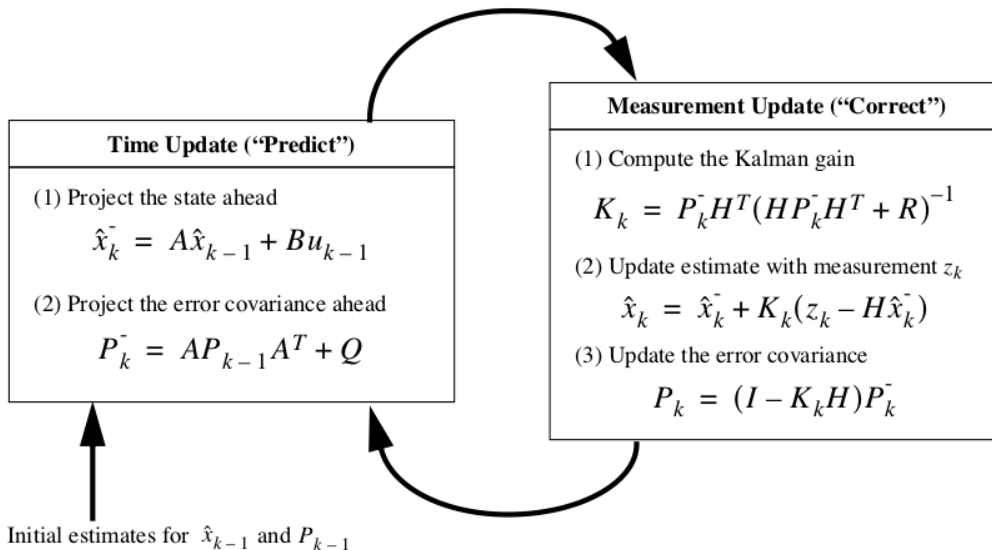
Transition

Uncertainty

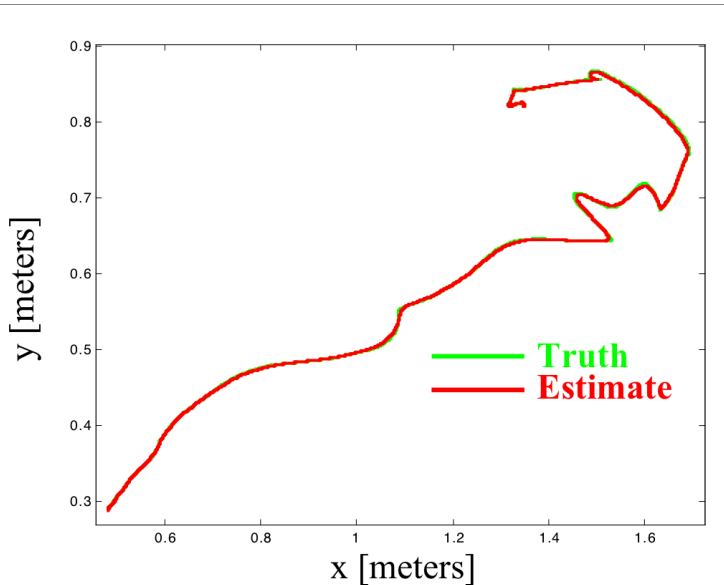
CORRECT



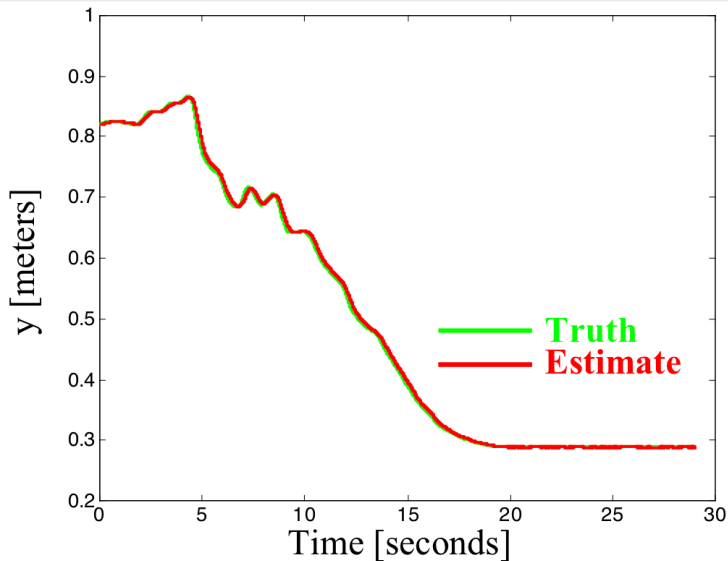
Summary



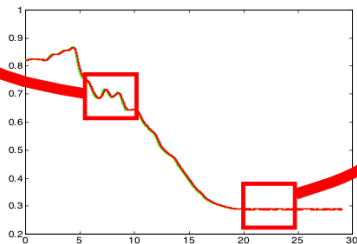
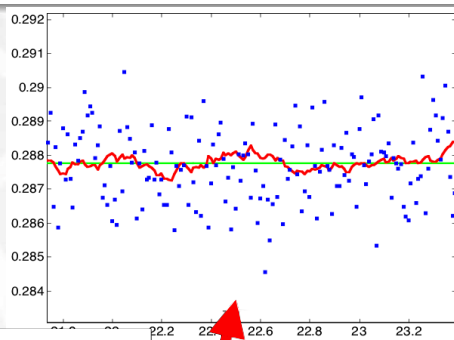
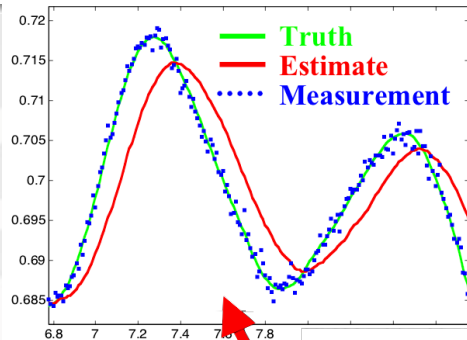
Results: XY Track



Y Track: Moving then Still



Motion-Dependent Performance





Process Model

- Describes how the state changes over time
- The state for the first example was scalar
- The process model was “nothing changes”

A better model might be

- State is a 2-vector [$position, velocity$]
- $position_{n+1} = position_n + velocity_n \times time$
- $velocity_{n+1} = velocity_n$

Example: 2D Position-Velocity (PV model)

Process Model (PV):

State Transition

$$\begin{bmatrix} 1 & 0 & dt & 0 \\ 0 & 1 & 0 & dt \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

State

$$\begin{bmatrix} x \\ y \\ \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix}$$

$$\bar{x}_k = A\bar{x}_{k-1} + \bar{w}_{k-1}$$

Measurement Model (PV):

Measurement Matrix

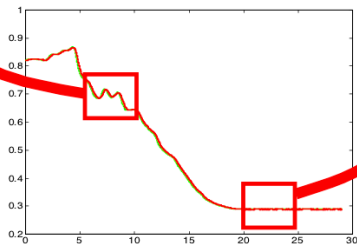
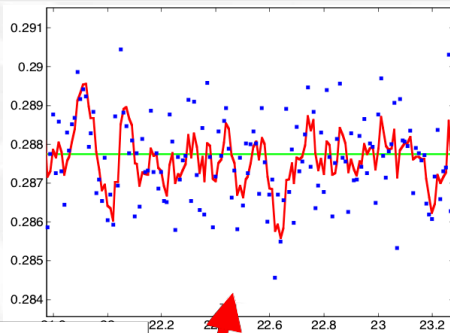
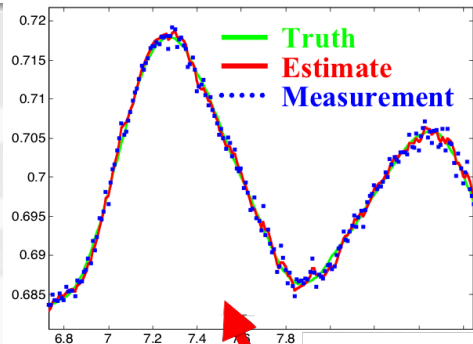
$$\begin{bmatrix} H_x & 0 & 0 & 0 \\ 0 & H_y & 0 & 0 \end{bmatrix}$$

State

$$\begin{bmatrix} x \\ y \\ \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix}$$

$$\bar{z}_k = H\bar{x}_k + \bar{v}_k$$

Motion-Dependent Performance





Action and perception updates

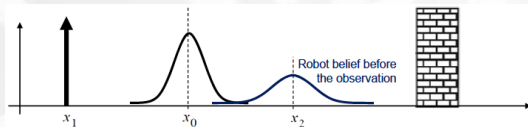
In robot localization, we distinguish two update steps:

Action and perception updates

In robot localization, we distinguish two update steps:

① **Action** update:

- the robot moves and estimates its position through its proprioceptive sensors. During this step, the robot uncertainty grows.

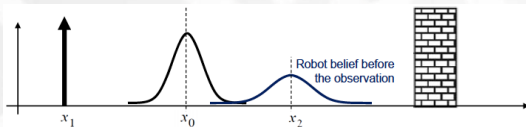


Action and perception updates

In robot localization, we distinguish two update steps:

1 Action update:

- the robot moves and estimates its position through its proprioceptive sensors. During this step, the robot uncertainty grows.



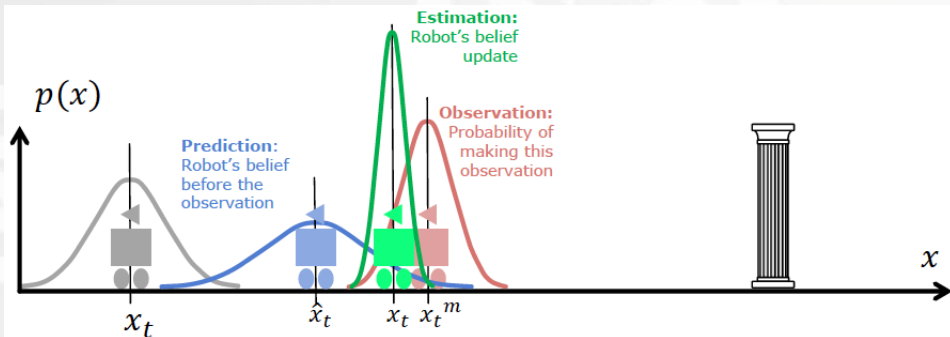
2 Perception update:

- the robot makes an observation using its exteroceptive sensors and correct its position by opportunely combining its belief before the observation with the probability of making exactly that observation. During this step, the robot **uncertainty shrinks**.



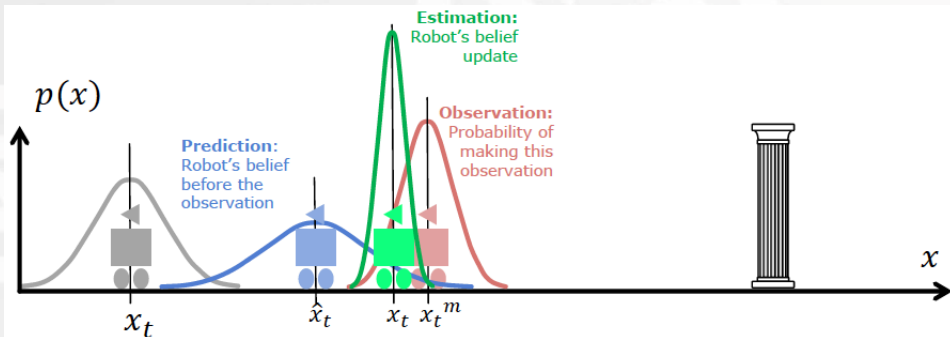
Kalman Filter Localization | in summery

① Prediction (ACT) based on previous estimate and odometry



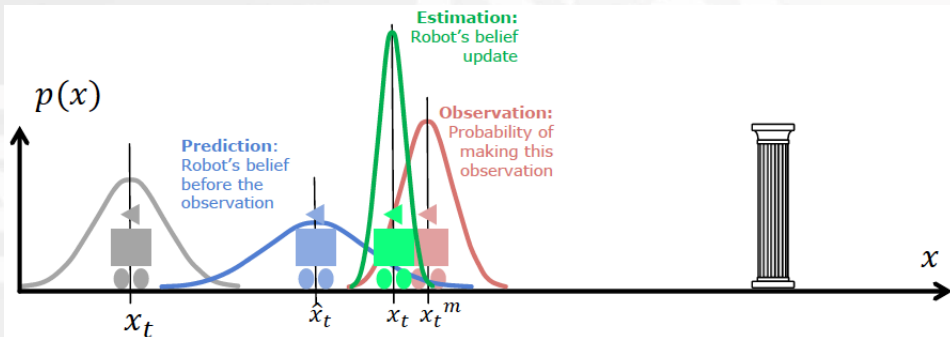
Kalman Filter Localization | in summery

- 1 Prediction (ACT) based on previous estimate and odometry
- 2 Observation (SEE) with on-board sensors



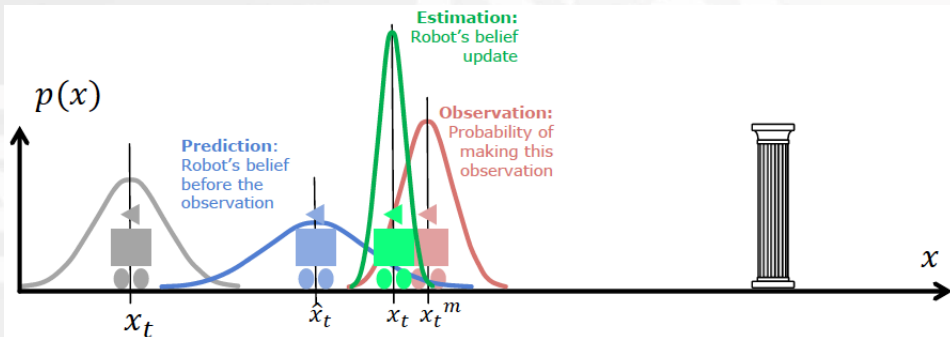
Kalman Filter Localization | in summery

- 1 Prediction (ACT) based on previous estimate and odometry
- 2 Observation (SEE) with on-board sensors
- 3 Measurement prediction based on prediction and map



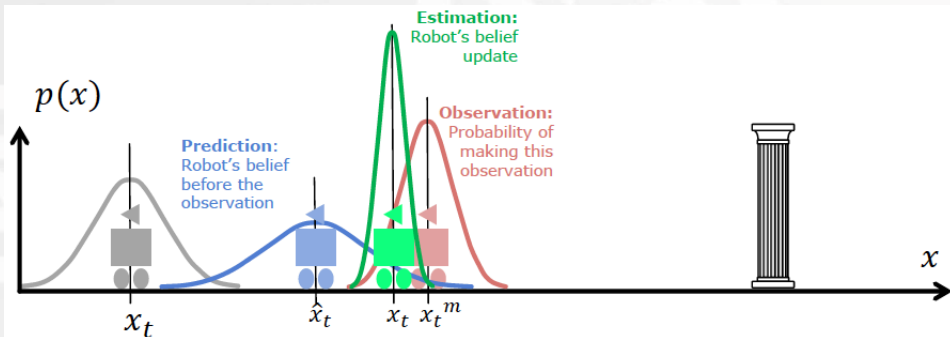
Kalman Filter Localization | in summery

- 1 Prediction (ACT) based on previous estimate and odometry
- 2 Observation (SEE) with on-board sensors
- 3 Measurement prediction based on prediction and map
- 4 Matching of observation and map



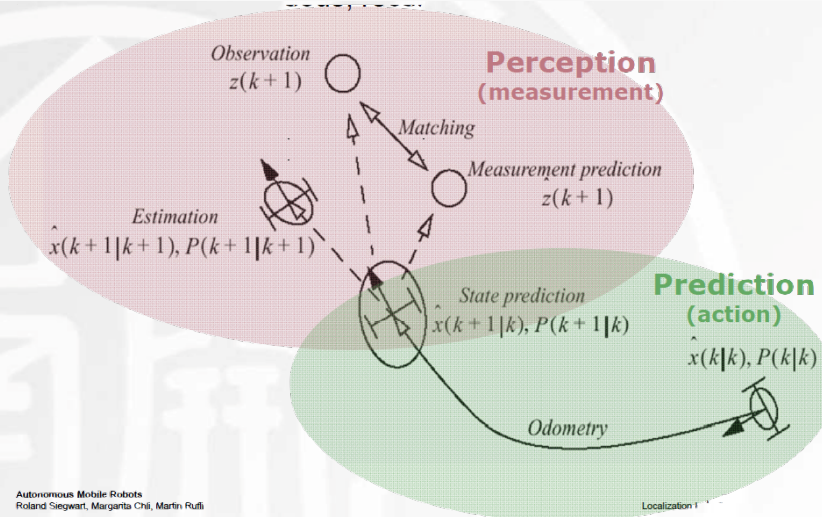
Kalman Filter Localization | in summery

- 1 Prediction (ACT) based on previous estimate and odometry
- 2 Observation (SEE) with on-board sensors
- 3 Measurement prediction based on prediction and map
- 4 Matching of observation and map
- 5 Estimation $\rightarrow$ position update (posteriori position) test



Localization: Probabilistic Position Estimation

(Kalman Filter: continuous, recursive and very compact)



Autonomous Mobile Robots
Roland Siegwart, Margarita Chli, Martin Rufli

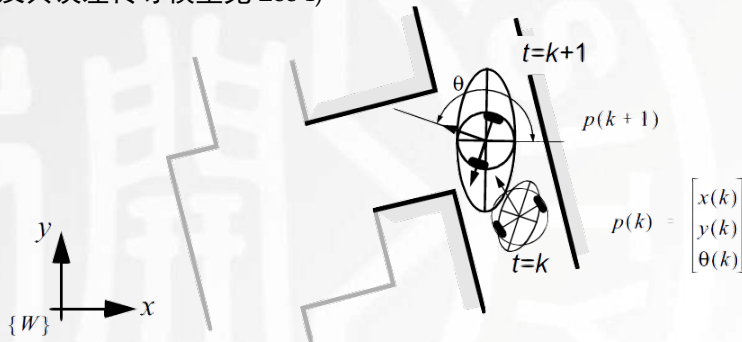
Robot Position Prediction

In a first step, the robots position $\hat{x}_t$ at time step t is predicted based on its old location (time step $t - 1$) and its movement due to the control input u_t :

$$\hat{x}_t = f(x_{t-1}, u_t)$$

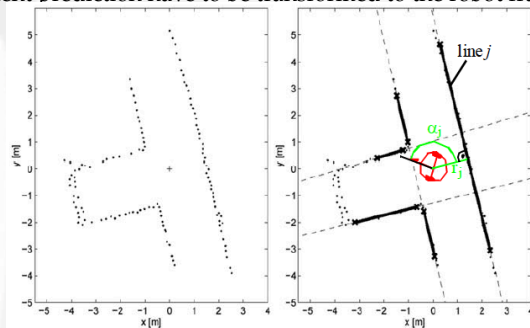
$$\hat{P}_t = F_x \cdot P_{t-1} \cdot F_x^T + F_u \cdot Q_t \cdot F_u^T$$

(里程计模型及其误差传导模型见 Lec 4)



Observation

- The second step is to obtain the observation Z (measurements) from the robot's sensors at the new location
- The observation usually consists of a set n_0 of single observations z_j extracted from the different sensors signals. It represents features like lines, doors or any kind of landmarks.
- The parameters of the targets are **usually observed in the sensor frame $\{S\}$** .
 - Therefore the observations have to be transformed to the world frame $\{W\}$ or
 - the measurement prediction have to be transformed to the robot frame $\{R\}$.





Measurement Prediction

- In the next step we use the predicted robot position $\hat{x}_t$ and the features m^j in the map M to generate multiple predicted observations $\hat{z}_t^j$
- They have to be transformed into the sensor frame

$$\hat{z}_t^j = h^j(\hat{x}_t, m^j)$$

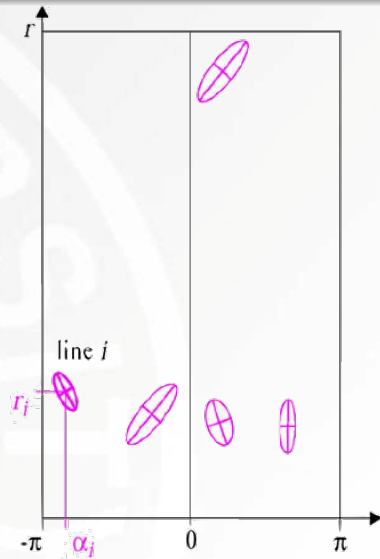
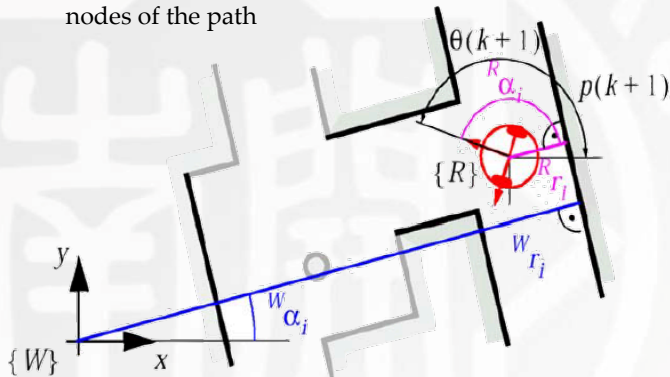
- We can now define the measurement prediction as the set containing all n^j predicted observations

$$\hat{Z} = \{\hat{z}_j | 1 \leq j \leq n_j\}$$

- The function h^j is mainly the coordinate transformation between the world $\{W\}$ frame and the sensor/robot frame $\{R\}$

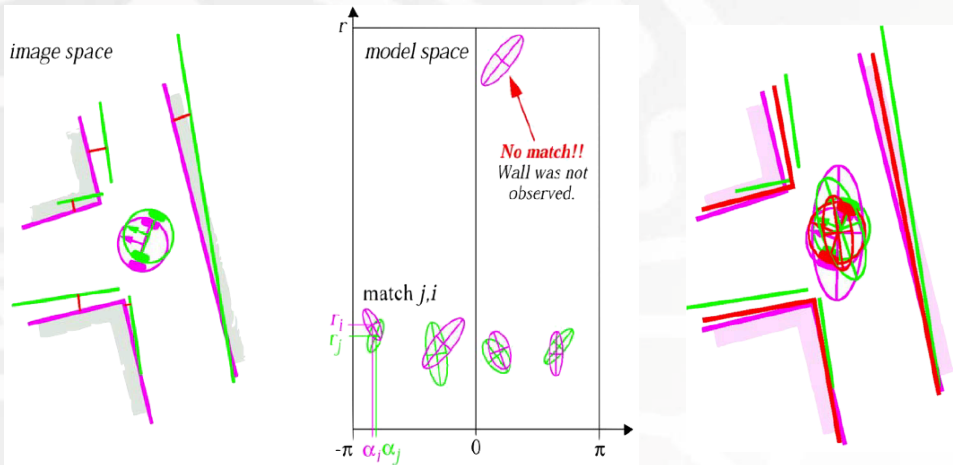
Measurement Prediction

- For prediction, only the walls that are in the field of view of the robot are selected.
- This is done by linking the individual lines to the nodes of the path



Estimation

- Kalman filter estimation of the new robot position x_t :
 - By fusing the prediction of robot position (magenta) with the innovation gained by the measurements (green) we get the updated estimate of the robot position (red)





基于卡尔曼滤波定位算法流程 (1)——先验估计

① 预测模型（先验估计）：里程计模型（世界坐标系下）

$$\begin{cases} x_{k+1} = x_k + \Delta s \cos(\theta_k + \frac{\Delta\theta}{2}) \\ y_{k+1} = y_k + \Delta s \sin(\theta_k + \frac{\Delta\theta}{2}) \\ \theta_{k+1} = \theta_k + \Delta\theta \end{cases}$$

其中，码盘读数为：

$$\begin{cases} \Delta s = \frac{r}{2}(\Delta\phi_R + \Delta\phi_L) \\ \Delta\theta = \frac{r}{2d}(\Delta\phi_R - \Delta\phi_L) \end{cases}$$

基于卡尔曼滤波定位算法流程 (2)——先验估计误差

② 先验估计误差模型：里程计误差传导（非线性）

$$\sum_{p'} = \nabla_p f \cdot \sum_p \cdot \nabla_p f^T + \nabla_{rl} f \cdot \sum_{\Delta} \cdot \nabla_{rl} f^T$$

其中：

$$F_p = \nabla_p f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial \theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\Delta s \sin(\theta_k + \frac{\Delta \theta}{2}) \\ 0 & 1 & \Delta s \cos(\theta_k + \frac{\Delta \theta}{2}) \\ 0 & 0 & 1 \end{bmatrix}$$

$$F_{\Delta} = \nabla_{rl} f = \begin{bmatrix} \frac{\partial f}{\partial \Delta \phi_R} & \frac{\partial f}{\partial \Delta \phi_L} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{r}{2} \cos(\theta_k + \frac{\Delta \theta}{2}) - \frac{r}{4d} \Delta s \sin(\theta_k + \frac{\Delta \theta}{2}) & \frac{r}{2} \cos(\theta_k + \frac{\Delta \theta}{2}) + \frac{r}{4d} \Delta s \sin(\theta_k + \frac{\Delta \theta}{2}) \\ \frac{r}{2} \sin(\theta_k + \frac{\Delta \theta}{2}) + \frac{r}{4d} \Delta s \cos(\theta_k + \frac{\Delta \theta}{2}) & \frac{r}{2} \sin(\theta_k + \frac{\Delta \theta}{2}) - \frac{r}{4d} \Delta s \cos(\theta_k + \frac{\Delta \theta}{2}) \end{bmatrix}$$

基于卡尔曼滤波定位算法流程 (3)——观测模型

③ 观测模型：激光定位算法（ICP：确定匹配关系，SVD：确定机器人位姿）

给定两个在 2 维空间中对应的点集合 $P = \{p_1, p_2, \dots, p_n\}$ 和 $Q = \{q_1, q_2, \dots, q_n\}$ ：

- 加权平均：

$$\hat{p} = \frac{\sum_{i=1}^n w_i p_i}{\sum_{i=1}^n w_i} \quad \hat{q} = \frac{\sum_{i=1}^n w_i q_i}{\sum_{i=1}^n w_i}$$

其中 w_i 为权重参数，可根据激光测量距离取 $w_i = 1/\sigma_i(\rho_i)$ 。

- 去中心化：

$$x_i := p_i - \hat{p} \quad y_i := q_i - \hat{q}$$

构造矩阵 $W = \text{diag}(w_1, \dots, w_n)$, $X = [x_1, x_2, \dots, x_n]$, $Y = [y_1, y_2, \dots, y_n] \in R^{2 \times n}$.

- 构造 SVD 矩阵：令 $S = XWY^T = U\Sigma V^T$ ，由 SVD 原理，求取 U, V 单位正交阵。
- 求取旋转矩阵 R ：
$$R = VU^T$$
- 求取平移矩阵 t ：
$$t = \hat{q} - R\hat{p}$$



基于卡尔曼滤波定位算法流程 (4)——观测误差模型

④ 观测误差模型：SVD 定位误差模型

最小的匹配误差为：

$$\arg \min \sum_{i=1}^n w_i \| (Rp_i + t) - q_i \|^2 = \sum_{i=1}^n w_i x_i^T x_i + \sum_{i=1}^n w_i y_i^T y_i - 2 \sum_{i=1}^n \sigma_i$$

其中，

- w_i 为权重参数，可根据激光测量距离取 $w_i = 1/\sigma_i(\rho_i)$ ，
- ρ_i 为激光测距方差，可对激光雷达离线标定获得。
- $x_i := p_i - \hat{p}$ 为地图特征去中心化后的点。
- $y_i := q_i - \hat{q}$ 为激光雷达坐标系下识别的特征去中心化后的点。

基于卡尔曼滤波定位算法流程 (5)——后验估计

⑤ 基于观测的后验估计（对先验估计的修正）：

- 卡尔曼增益 K ：

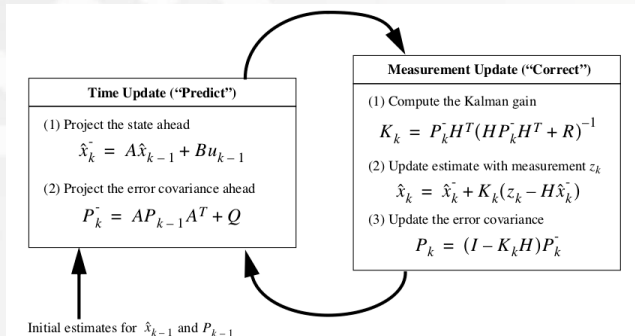
$$K = P_k^- H^T (H P_k^- H^T + R)^{-1}$$

- 后验估计 $\bar{x}_k$ ：

$$\bar{x}_k = \bar{x}_k^- + K (\bar{z}_k - H \bar{x}_k^-)$$

- 后验估计误差 P_k ：

$$P_k = (I - KH)P_k^-$$



- 激光测量频率小于里程计测量频率，如何处理？
- 激光测量有可能仅能得到机器人部分状态，如何处理？
- 该方法也可融合诸如 IMU 等其他传感器信息，如何融合？



基于卡尔曼滤波定位算法总结

- 基于卡尔曼滤波定位算法仅能处理位置跟踪问题；
 - 根据里程计推演信息，可得到对机器人当前位姿的估计
 - 根据对机器人当前位姿估计，在地图中进行匹配搜索，避免 ICP 陷入局部最优，导致定位失败。
 - 激光定位精度一般在2-3cm，使用基于卡尔曼滤波定位算法融合里程计后，定位精度可达5mm左右。
- 严重依赖于匹配精度，如果出现匹配错误，则机器人定位失败
- 无法判断定位失败
- 无法从定位失败中恢复
- 收敛速度受初始状态误差，以及协方差阵的精度影响较大
- 无法进行全局定位

第五章 移动机器人感知技术

Kalman Filter Based Localization

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